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JEE MAIN · QUESTION PAPER

JEE Main 2022 (Online) 29th July Morning Shift

JEE Main Question Paper with Answer Key

29 Jul 2022	90	14	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	24 questions
Mathematics	23 questions
Physics	29 questions
Out of syllabus	14 questions
Answer key	at the end

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Chemistry 24 questions

Q1. MCQ (Single Correct Answer)

Which of the following pair of molecules contains an odd electron molecule and an expanded octet molecule?

(A) BCl_3 and SF_6
(B) NO and H_2SO_4
(C) SF_6 and H_2SO_4
(D) BCl_3 and NO

Open solution

Q2. MCQ (Single Correct Answer)

$N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$

$20\ g \quad 5\ g$

Consider the above reaction, the limiting reagent of the reaction and number of moles of NH_3 formed respectively are :

(A) $H_2, 1.42$ moles
(B) $H_2, 0.71$ moles
(C) $N_2, 1.42$ moles
(D) $N_2, 0.71$ moles

Open solution

Q3. MCQ (Single Correct Answer)

The first ionization enthalpy of Na, Mg and Si, respectively, are : 496, 737 and 786 $kJ\ mol^{-1}$. The first ionization enthalpy ($kJ\ mol^{-1}$) of Al is :

(A) 487
(B) 768
(C) 577
(D) 856

Open solution

Q4. MCQ (Single Correct Answer)

The reaction of zinc with excess of aqueous alkali, evolves hydrogen gas and gives :

(A) $Zn(OH)_2$
(B) ZnO
(C) $[Zn(OH)_4]^{2-}$
(D) $[ZnO_2]^{2-}$

Open solution

Q5. MCQ (Single Correct Answer)

Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are :

(A) 0, 1, 2 and 2
(B) 2, 1, 2 and 0
(C) 1, 2, 2 and 0
(D) 2, 1, 0 and 2

Open solution

Q6. MCQ (Single Correct Answer)

In following pairs, the one in which both transition metal ions are colourless is :

(A) Sc^{3+}, Zn^{2+}
(B) Ti^{4+}, Cu^{2+}
(C) V^{2+}, Ti^{3+}
(D) Zn^{2+}, Mn^{2+}

Open solution

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Q7. MCQ (Single Correct Answer)

In neutral or faintly alkaline medium, $KMnO_4$ being a powerful oxidant can oxidize, thiosulphate almost quantitatively, to sulphate. In this reaction overall change in oxidation state of manganese will be :

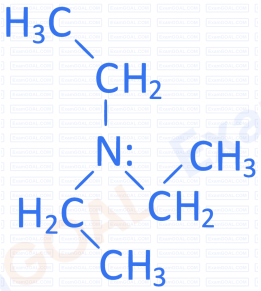
Open solution

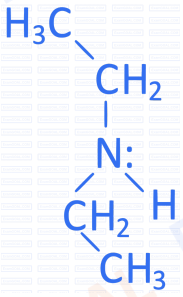
(A) 5 (B) 1 (C) 0 (D) 3

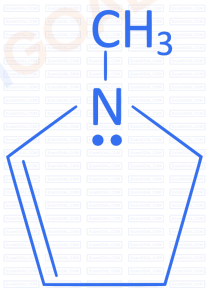
Q8. MCQ (Single Correct Answer)

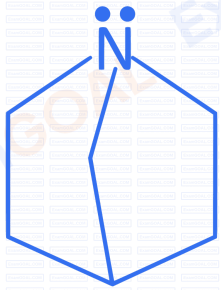
Which among the following is the strongest Bronsted base?

Open solution

(A) 

(B) 

(C) 

(D) 

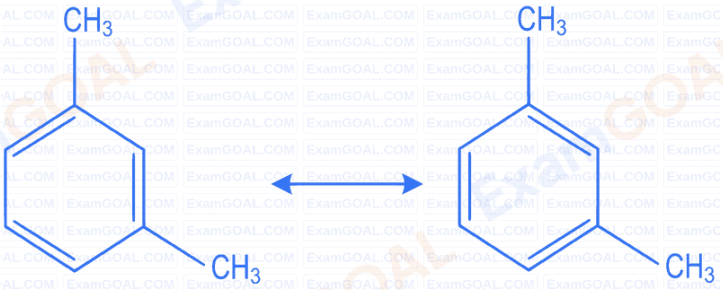
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Q9. MCQ (Single Correct Answer)

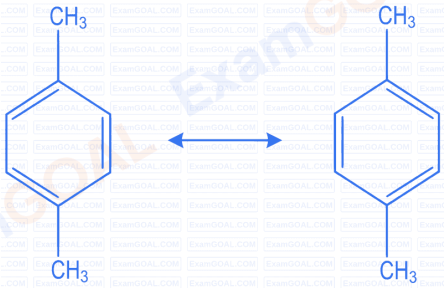
Open solution

Which among the following pairs of the structures will give different products on ozonolysis? (Consider the double bonds in the structures are rigid and not delocalized.)

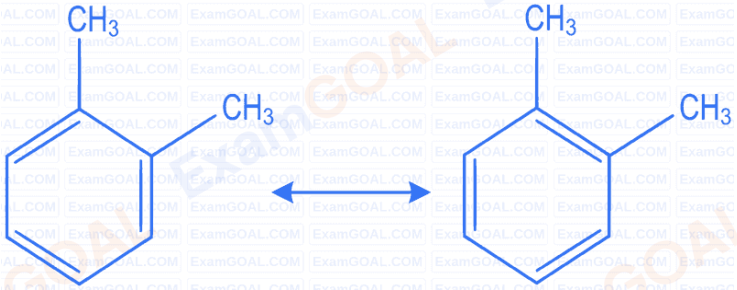
(A)



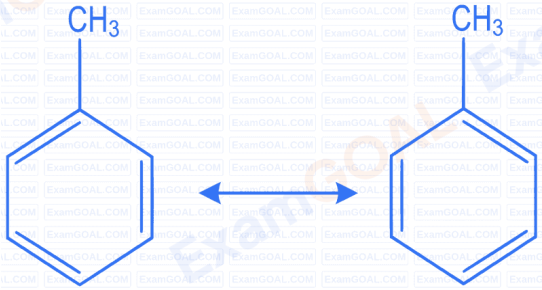
(B)



(C)



(D)



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Q10. MCQ (Single Correct Answer)



Considering the above reactions, the compound 'A' and compound 'B' respectively are :

(A) 

(B) 

(C) 

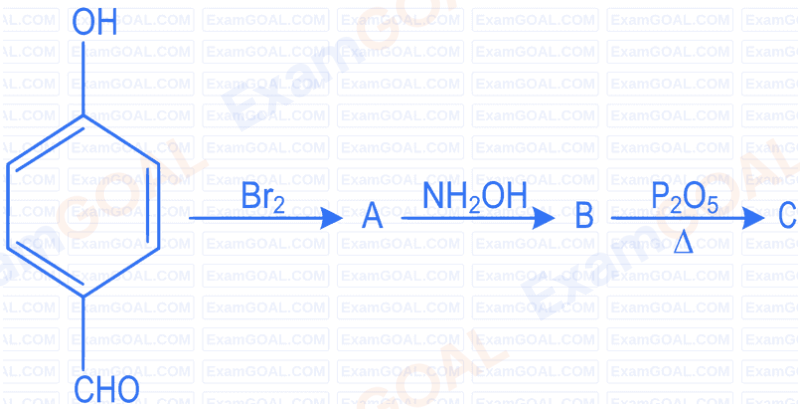
(D) 

Open solution

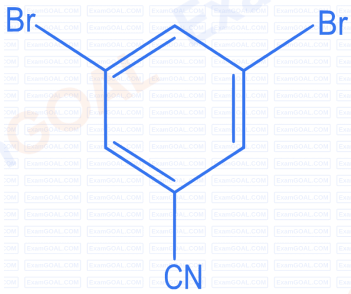
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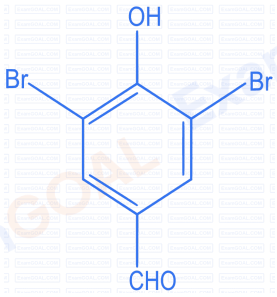
Q11. MCQ (Single Correct Answer)

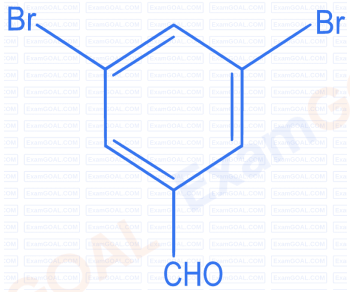
Open solution

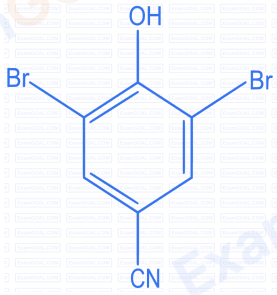


Consider the above reaction sequence, the Product 'C' is :

(A) 

(B) 

(C) 

(D) 

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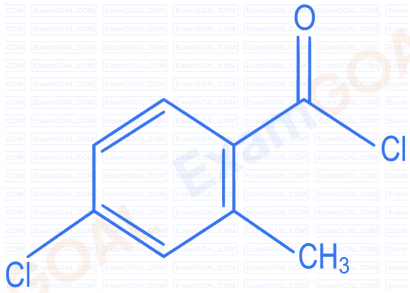
Q12. MCQ (Single Correct Answer)

Open solution

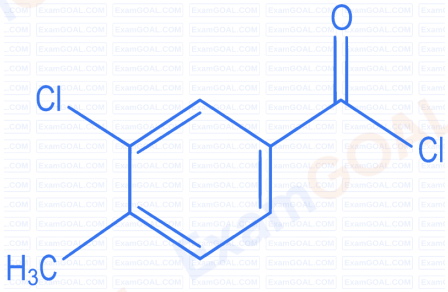


Consider the above reaction, the compound 'A' is :

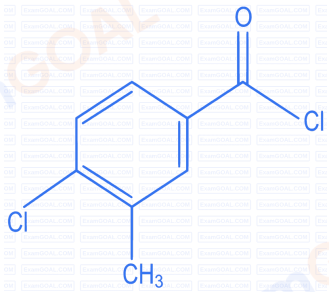
(A)



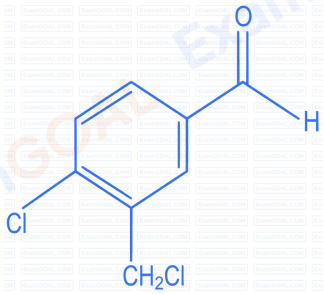
(B)



(C)

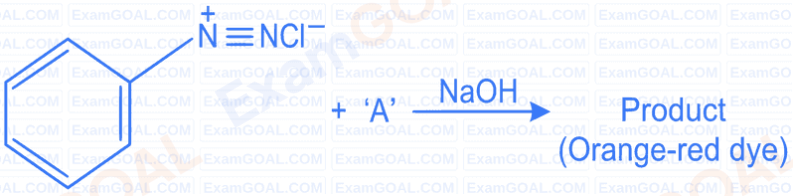


(D)



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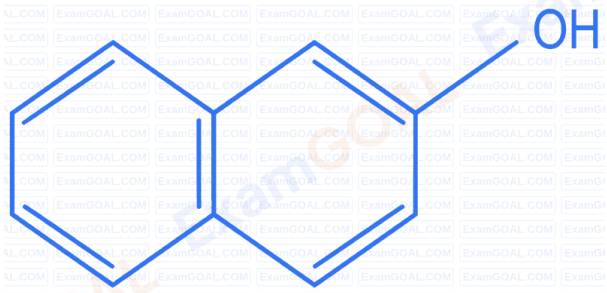
Q13. MCQ (Single Correct Answer)



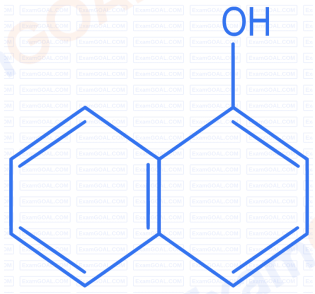
Open solution

Which among the following represents reagent 'A'?

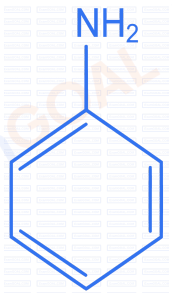
(A)



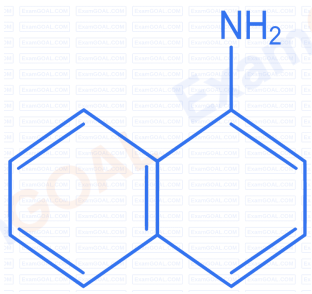
(B)



(C)



(D)

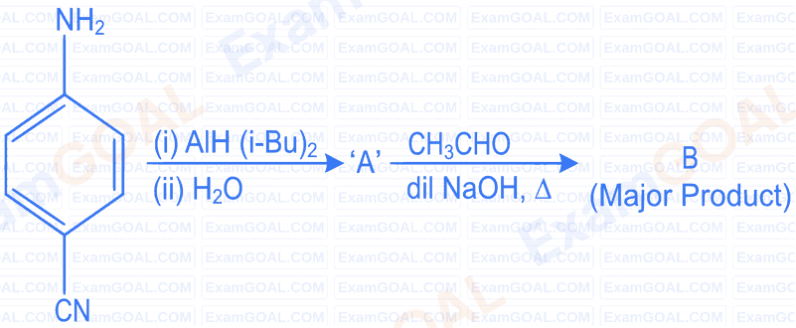


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Q14. MCQ (Single Correct Answer)

Open solution

Consider the following reaction sequence:



The product 'B' is :

(A)



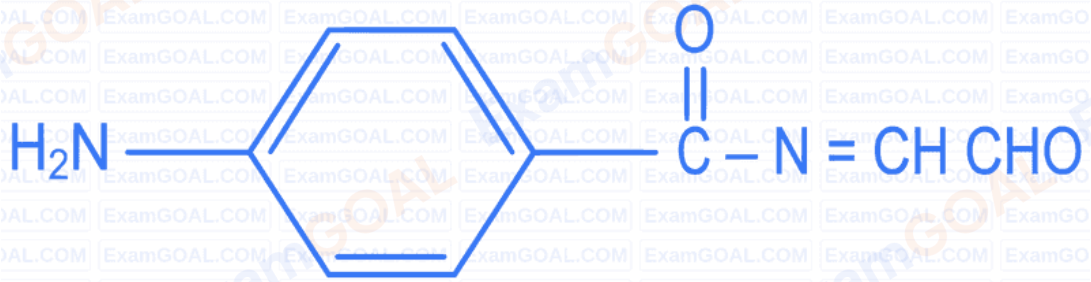
(B)



(C)



(D)

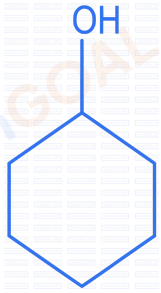


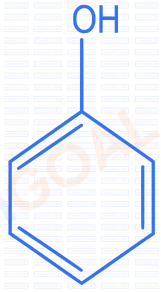
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Q15. MCQ (Single Correct Answer)

Open solution

A compound 'X' is acidic and it is soluble in NaOH solution, but insoluble in $NaHCO_3$ solution. Compound 'X' also gives violet colour with neutral $FeCl_3$ solution. The compound 'X' is :

(A) 

(B) 

(C) 

(D) 

Q16. Numerical

Open solution

Resistance of a conductivity cell (cell constant $129\ \mathrm{m}^{-1}$) filled with $74.5\ \mathrm{ppm}$ solution of KCl is $100\ \Omega$ (labelled as solution 1). When the same cell is filled with KCl solution of $149\ \mathrm{ppm}$, the resistance is $50\ \Omega$ (labelled as solution 2). The ratio of molar conductivity of solution 1 and solution 2 is i.e. $\frac{\Lambda_1}{\Lambda_2} = x \times 10^{-3}$. The value of x is _____. (Nearest integer)

Given, molar mass of KCl is $74.5\ \mathrm{g\ mol}^{-1}$.

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Q17. Numerical

The minimum uncertainty in the speed of an electron in an one dimensional region of length $2a_o$ (Where a_o = Bohr radius $52.9\,pm$) is _____ $km\,s^{-1}$.

(Given : Mass of electron = 9.1×10^{-31} kg, Planck's constant h = 6.63×10^{-34} Js)

Open solution

Q18. Numerical

When 600 mL of 0.2 M HNO_3 is mixed with $400\,mL$ of 0.1 M NaOH solution in a flask, the rise in temperature of the flask is _____ $\times 10^{-2}\,^{\circ}C$.

(Enthalpy of neutralisation $= 57\,kJ\,mol^{-1}$ and Specific heat of water $= 4.2\,JK^{-1}\,g^{-1}$) (Neglect heat capacity of flask)

Open solution

Q19. Numerical

If O_2 gas is bubbled through water at $303\,K$, the number of millimoles of O_2 gas that dissolve in 1 litre of water is _____. (Nearest Integer)

(Given : Henry's Law constant for O_2 at $303\,K$ is $46.82\,k$ bar and partial pressure of $O_2 = 0.920$ bar)

(Assume solubility of O_2 in water is too small, nearly negligible)

Open solution

Q20. Numerical

If the solubility product of PbS is 8×10^{-28}, then the solubility of PbS in pure water at 298 K is x $\times$ 10^{-16} mol L^{-1}. The value of x is _____. (Nearest Integer)

[Given : $\sqrt{2}$ = 1.41]

Open solution

Q21. Numerical

The reaction between X and Y is first order with respect to X and zero order with respect to Y.

Experiment	$\frac{[X]}{mol\,L^{-1}}$	$\frac{[Y]}{mol\,L^{-1}}$	$\frac{Initial\ rate}{mol\,L^{-1}\,min^{-1}}$
I	0.1	0.1	2×10^{-3}
II	L	0.2	4×10^{-3}
III	0.4	0.4	$M \times 10^{-3}$
IV	0.1	0.2	2×10^{-3}

Examine the data of table and calculate ratio of numerical values of M and L. (Nearest Integer)

Open solution

Q22. Numerical

In a linear tetrapeptide (Constituted with different amino acids), (number of amino acids) $-$ (number of peptide bonds) is _____.

Open solution

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Q23. Numerical

In bromination of Propyne, with Bromine, 1, 1, 2, 2-tetrabromopropane is obtained in 27% yield. The amount of 1, 1, 2, 2-tetrabromopropane obtained from 1 g of Bromine in this reaction is _____ $\times 10^{-1}$ g. (Nearest integer)

(Molar Mass : Bromine = 80 g/mol)

Open solution

Q24. Numerical

$[Fe(CN)_6]^{3-}$ should be an inner orbital complex. Ignoring the pairing energy, the value of crystal field stabilization energy for this complex is $(-)$ _____ Δ_0. (Nearest integer)

Open solution

Mathematics 23 questions

Q25. MCQ (Single Correct Answer)

Let R be a relation from the set $\{1, 2, 3, \ldots, 60\}$ to itself such that $R = \{(a, b) : b = pq$, where $p, q \geq 3$ are prime numbers$\}$. Then, the number of elements in R is :

(A) 600 (B) 660 (C) 540 (D) 720

Open solution

Q26. MCQ (Single Correct Answer)

If $z = 2 + 3i$, then $z^5 + (\bar{z})^5$ is equal to :

(A) 244 (B) 224 (C) 245 (D) 265

Open solution

Q27. MCQ (Single Correct Answer)

Let A and B be two 3×3 non-zero real matrices such that AB is a zero matrix. Then

(A) the system of linear equations $AX = 0$ has a unique solution

(B) the system of linear equations $AX = 0$ has infinitely many solutions

(C) B is an invertible matrix

(D) $\mathrm{adj(A)}$ is an invertible matrix

Open solution

Q28. MCQ (Single Correct Answer)

If $\frac{1}{(20-a)(40-a)} + \frac{1}{(40-a)(60-a)} + \ldots + \frac{1}{(180-a)(200-a)} = \frac{1}{256}$, then the maximum value of a is :

(A) 198 (B) 202 (C) 212 (D) 218

Open solution

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Q29. MCQ (Single Correct Answer)

If $\lim\limits_{x\to 0}\frac{\alpha e^{x}+\beta e^{-x}+\gamma\sin x}{x\sin^2 x}=\frac{2}{3}$, where $\alpha,\beta,\gamma\in\mathbf{R}$, then which of the following is NOT correct?

(A) $\alpha^2+\beta^2+\gamma^2=6$

(B) $\alpha\beta+\beta\gamma+\gamma\alpha+1=0$

(C) $\alpha\beta^2+\beta\gamma^2+\gamma\alpha^2+3=0$

(D) $\alpha^2-\beta^2+\gamma^2=4$

Open solution

Q30. MCQ (Single Correct Answer)

The integral $\int\limits_0^{\frac{\pi}{2}}\frac{1}{3+2\sin x+\cos x}\,dx$ is equal to :

(A) $\tan^{-1}(2)$

(B) $\tan^{-1}(2)-\frac{\pi}{4}$

(C) $\frac{1}{2}\tan^{-1}(2)-\frac{\pi}{8}$

(D) $\frac{1}{2}$

Open solution

Q31. MCQ (Single Correct Answer)

Let the solution curve $y=y(x)$ of the differential equation $\left(1+e^{2x}\right)\left(\frac{dy}{dx}+y\right)=1$ pass through the point $\left(0,\frac{\pi}{2}\right)$. Then, $\lim\limits_{x\to\infty}e^{x}y(x)$ is equal to :

(A) $\frac{\pi}{4}$

(B) $\frac{3\pi}{4}$

(C) $\frac{\pi}{2}$

(D) $\frac{3\pi}{2}$

Open solution

Q32. MCQ (Single Correct Answer)

Let a line L pass through the point of intersection of the lines $bx+10y-8=0$ and $2x-3y=0$, $b\in\mathbf{R}-\left\{\frac{4}{3}\right\}$. If the line L also passes through the point $(1,1)$ and touches the circle $17\left(x^2+y^2\right)=16$, then the eccentricity of the ellipse $\frac{x^2}{5}+\frac{y^2}{b^2}=1$ is :

(A) $\frac{2}{\sqrt{5}}$

(B) $\sqrt{\frac{3}{5}}$

(C) $\frac{1}{\sqrt{5}}$

(D) $\sqrt{\frac{2}{5}}$

Open solution

Q33. MCQ (Single Correct Answer)

Let the circumcentre of a triangle with vertices A(a, 3), B(b, 5) and C(a, b), ab > 0 be P(1,1). If the line AP intersects the line BC at the point Q(k_1,k_2), then k_1+k_2 is equal to :

(A) 2

(B) $\frac{4}{7}$

(C) $\frac{2}{7}$

(D) 4

Open solution

Q34. MCQ (Single Correct Answer)

Let $\hat{a}$ and $\hat{b}$ be two unit vectors such that the angle between them is $\frac{\pi}{4}$. If θ is the angle between the vectors $(\hat{a}+\hat{b})$ and $(\hat{a}+2\hat{b}+2(\hat{a}\times\hat{b}))$, then the value of $164\cos^2\theta$ is equal to :

(A) $90+27\sqrt{2}$

(B) $45+18\sqrt{2}$

(C) $90+3\sqrt{2}$

(D) $54+90\sqrt{2}$

Open solution

Q35. MCQ (Single Correct Answer)

If $f(\alpha)=\int\limits_1^{\alpha}\frac{\log_{10}t}{1+t}dt,\alpha>0$, then $f\left(e^3\right)+f\left(e^{-3}\right)$ is equal to :

(A) 9

(B) $\frac{9}{2}$

(C) $\frac{9}{\log_e(10)}$

(D) $\frac{9}{2\log_e(10)}$

Open solution

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Q36. MCQ (Single Correct Answer)

The area of the region

$\left\{(x, y) : |x-1| \leq y \leq \sqrt{5-x^2}\right\}$ is equal to :

(A) $\frac{5}{2}\sin^{-1}\left(\frac{3}{5}\right) - \frac{1}{2}$

(B) $\frac{5\pi}{4} - \frac{3}{2}$

(C) $\frac{3\pi}{4} + \frac{3}{2}$

(D) $\frac{5\pi}{4} - \frac{1}{2}$

Open solution

Q37. MCQ (Single Correct Answer)

Let the focal chord of the parabola $\mathrm{P} : y^2 = 4x$ along the line $\mathrm{L} : y = \mathrm{m}x + \mathrm{c}, \mathrm{m} > 0$ meet the parabola at the points M and N. Let the line L be a tangent to the hyperbola $\mathrm{H} : x^2 - y^2 = 4$. If O is the vertex of P and F is the focus of H on the positive x-axis, then the area of the quadrilateral OMFN is :

(A) $2\sqrt{6}$

(B) $2\sqrt{14}$

(C) $4\sqrt{6}$

(D) $4\sqrt{14}$

Open solution

Q38. MCQ (Single Correct Answer)

The number of points, where the function $f : \mathbf{R} \to \mathbf{R}$,

$f(x) = |x-1|\cos|x-2|\sin|x-1| + (x-3)\left|x^2 - 5x + 4\right|$, is NOT differentiable, is :

(A) 1

(B) 2

(C) 3

(D) 4

Open solution

Q39. MCQ (Single Correct Answer)

Let $S = \{1, 2, 3, \ldots, 2022\}$. Then the probability, that a randomly chosen number n from the set S such that $\mathrm{HCF}\,(\mathrm{n}, 2022) = 1$, is :

(A) $\frac{128}{1011}$

(B) $\frac{166}{1011}$

(C) $\frac{127}{337}$

(D) $\frac{112}{337}$

Open solution

Q40. MCQ (Single Correct Answer)

Let $f(x) = 3^{(x^2-2)^3+4}, x \in \mathrm{R}$. Then which of the following statements are true?

$\mathrm{P} : x = 0$ is a point of local minima of f

$\mathrm{Q} : x = \sqrt{2}$ is a point of inflection of f

$R : f'$ is increasing for $x > \sqrt{2}$

(A) Only P and Q

(B) Only P and R

(C) Only Q and R

(D) All P, Q and R

Open solution

Q41. Numerical

Let the mean and the variance of 20 observations $x_1, x_2, \ldots, x_{20}$ be 15 and 9 , respectively. For $\alpha \in \mathbf{R}$, if the mean of $(x_1+\alpha)^2, (x_2+\alpha)^2, \ldots, (x_{20}+\alpha)^2$ is 178 , then the square of the maximum value of α is equal to _____.

Open solution

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Q42. Numerical

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\sum_{r=1}^{\infty} \frac{a_r}{2^r} = 4$, then $4a_2$ is equal to _____.

Open solution

Q43. Numerical

Let the ratio of the fifth term from the beginning to the fifth term from the end in the binomial expansion of $\left(\sqrt[4]{2} + \frac{1}{\sqrt[4]{3}}\right)^n$, in the increasing powers of $\frac{1}{\sqrt[4]{3}}$ be $\sqrt[4]{6} : 1$. If the sixth term from the beginning is $\frac{\alpha}{\sqrt[4]{3}}$, then α is equal to _____.

Open solution

Q44. Numerical

The number of matrices of order 3×3, whose entries are either 0 or 1 and the sum of all the entries is a prime number, is _____.

Open solution

Q45. Numerical

Let p and p + 2 be prime numbers and let

$$\Delta = \begin{vmatrix} p! & (p+1)! & (p+2)! \\ (p+1)! & (p+2)! & (p+3)! \\ (p+2)! & (p+3)! & (p+4)! \end{vmatrix}$$

Then the sum of the maximum values of α and β, such that p^{α} and $(p+2)^{\beta}$ divide Δ, is _____.

Open solution

Q46. Numerical

Let $S = \{4, 6, 9\}$ and $T = \{9, 10, 11, \ldots, 1000\}$. If $A = \{a_1 + a_2 + \ldots + a_k : k \in \mathbf{N}, a_1, a_2, a_3, \ldots, a_k \in S\}$, then the sum of all the elements in the set $T - A$ is equal to _____.

Open solution

Q47. Numerical

Let the mirror image of a circle $c_1 : x^2 + y^2 - 2x - 6y + \alpha = 0$ in line $y = x + 1$ be $c_2 : 5x^2 + 5y^2 + 10gx + 10fy + 38 = 0$. If r is the radius of circle c_2, then $\alpha + 6r^2$ is equal to _____.

Open solution

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Physics 29 questions

Q48. MCQ (Single Correct Answer)

Open solution

Given below are two statements : One is labelled as Assertion (A) and other is labelled as Reason (R).

Assertion (A) : Time period of oscillation of a liquid drop depends on surface tension (S), if density of the liquid is ρ and radius of the drop is r, then $\mathrm{T}=\mathrm{K}\sqrt{\rho \mathrm{r}^3/\mathrm{S}^{3/2}}$ is dimensionally correct, where K is dimensionless.

Reason (R) : Using dimensional analysis we get R.H.S. having different dimension than that of time period.

In the light of above statements, choose the correct answer from the options given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q49. MCQ (Single Correct Answer)

Open solution

A ball is thrown up vertically with a certain velocity so that, it reaches a maximum height h. Find the ratio of the times in which it is at height $\frac{h}{3}$ while going up and coming down respectively.

(A) $\frac{\sqrt{2}-1}{\sqrt{2}+1}$

(B) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

(C) $\frac{\sqrt{3}-1}{\sqrt{3}+1}$

(D) $\frac{1}{3}$

Q50. MCQ (Single Correct Answer)

Open solution

If $\mathrm{t}=\sqrt{x}+4$, then $\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)_{\mathrm{t}=4}$ is :

(A) 4

(B) zero

(C) 8

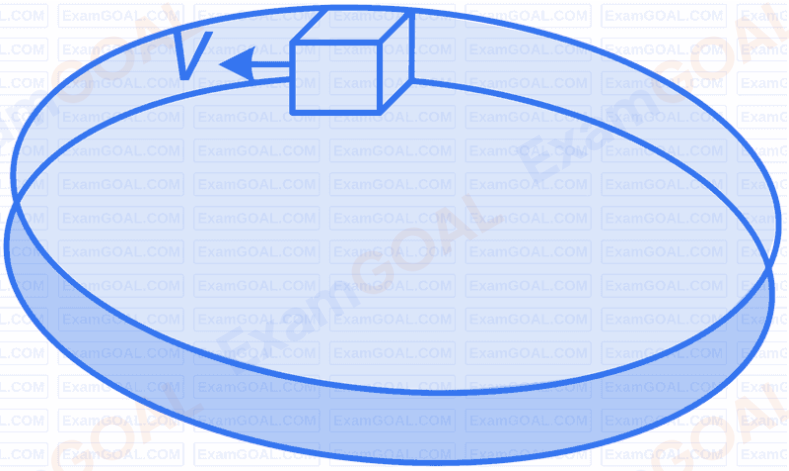
(D) 16

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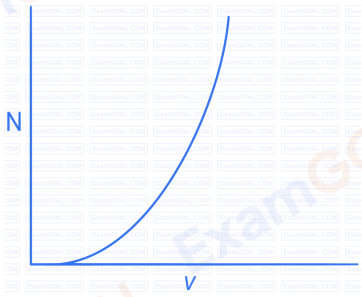
Q51. MCQ (Single Correct Answer)

Open solution

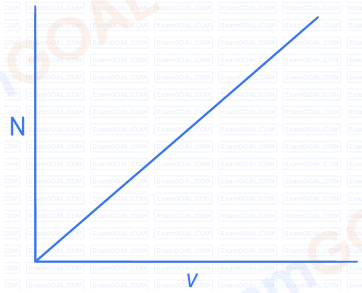
A smooth circular groove has a smooth vertical wall as shown in figure. A block of mass m moves against the wall with a speed v. Which of the following curve represents the correct relation between the normal reaction on the block by the wall (N) and speed of the block (v) ?



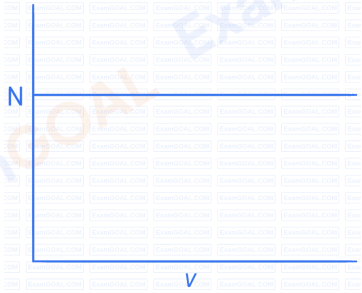
(A)



(B)

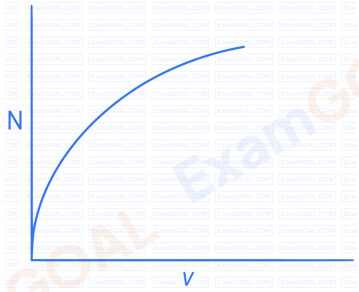


(C)



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(D)



Q52. MCQ (Single Correct Answer)

A ball is projected with kinetic energy E, at an angle of 60° to the horizontal. The kinetic energy of this ball at the highest point of its flight will become :

(A) Zero (B) $\frac{E}{2}$ (C) $\frac{E}{4}$ (D) E

Open solution

Q53. MCQ (Single Correct Answer)

Two bodies of mass $1\ \text{kg}$ and $3\ \text{kg}$ have position vectors $\hat{i}+2\hat{j}+\hat{k}$ and $-3\hat{i}-2\hat{j}+\hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

(A) $\hat{i}+2\hat{j}+\hat{k}$ (B) $-3\hat{i}-2\hat{j}+\hat{k}$ (C) $-2\hat{i}+2\hat{k}$ (D) $2\hat{i}-\hat{j}+2\hat{k}$

Open solution

Q54. MCQ (Single Correct Answer)

Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A): Clothes containing oil or grease stains cannot be cleaned by water wash.

Reason (R): Because the angle of contact between the oil/grease and water is obtuse.

In the light of the above statements, choose the correct answer from the option given below.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Open solution

Q55. MCQ (Single Correct Answer)

If the length of a wire is made double and radius is halved of its respective values. Then, the Young's modulus of the material of the wire will :

(A) remain same (B) become 8 times its initial value

(C) become $\frac{1}{4}$ of its initial value (D) become 4 times its initial value

Open solution

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Q56. MCQ (Single Correct Answer)

Open solution

The time period of oscillation of a simple pendulum of length L suspended from the roof of a vehicle, which moves without friction down an inclined plane of inclination α, is given by :

(A) $2\pi\sqrt{\mathrm{L}/(\mathrm{g}\cos\alpha)}$

(B) $2\pi\sqrt{\mathrm{L}/(\mathrm{g}\sin\alpha)}$

(C) $2\pi\sqrt{\mathrm{L}/\mathrm{g}}$

(D) $2\pi\sqrt{\mathrm{L}/(\mathrm{g}\tan\alpha)}$

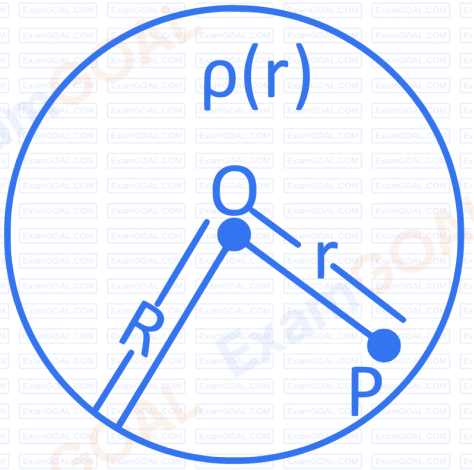
Q57. MCQ (Single Correct Answer)

Open solution

A spherically symmetric charge distribution is considered with charge density varying as

$$\rho(r)=\begin{cases}\rho_0\left(\frac{3}{4}-\frac{r}{R}\right) & \text{for } r\le R\\ \text{zero} & \text{for } r>R\end{cases}$$

Where, $r(r<R)$ is the distance from the centre O (as shown in figure). The electric field at point P will be:



(A) $\frac{\rho_0 r}{4\varepsilon_0}\left(\frac{3}{4}-\frac{r}{R}\right)$

(B) $\frac{\rho_0 r}{3\varepsilon_0}\left(\frac{3}{4}-\frac{r}{R}\right)$

(C) $\frac{\rho_0 r}{4\varepsilon_0}\left(1-\frac{r}{R}\right)$

(D) $\frac{\rho_0 r}{5\varepsilon_0}\left(1-\frac{r}{R}\right)$

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Q58. MCQ (Single Correct Answer)

Open solution

Given below are two statements.

Statement I : Electric potential is constant within and at the surface of each conductor.

Statement II : Electric field just outside a charged conductor is perpendicular to the surface of the conductor at every point.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q59. MCQ (Single Correct Answer)

Open solution

Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of these wires respectively, the effective conductivity of the combination is :

(A) $\frac{\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

(B) $\frac{2\sigma_1\sigma_2}{\sigma_1+\sigma_2}$

(C) $\frac{\sigma_1+\sigma_2}{2\sigma_1\sigma_2}$

(D) $\frac{\sigma_1+\sigma_2}{\sigma_1\sigma_2}$

Q60. MCQ (Single Correct Answer)

Open solution

An alternating emf $\mathrm{E} = 440\sin 100\pi\mathrm{t}$ is applied to a circuit containing an inductance of $\frac{\sqrt{2}}{\pi}\mathrm{H}$. If an a.c. ammeter is connected in the circuit, its reading will be :

(A) 4.4 A

(B) 1.55 A

(C) 2.2 A

(D) 3.11 A

Q61. MCQ (Single Correct Answer)

Open solution

A coil of inductance 1 H and resistance $100\,\Omega$ is connected to a battery of 6 V. Determine approximately :

(a) The time elapsed before the current acquires half of its steady - state value.

(b) The energy stored in the magnetic field associated with the coil at an instant 15 ms after the circuit is switched on. (Given $\ln 2 = 0.693, \mathrm{e}^{-3/2} = 0.25$)

(A) t = 10 ms; U = 2 mJ

(B) t = 10 ms; U = 1 mJ

(C) t = 7 ms; U = 1 mJ

(D) t = 7 ms; U = 2 mJ

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Q62. MCQ (Single Correct Answer)

Open solution

Match List - I with List - II :

	List - I		List - II
(a)	UV rays	(i)	Diagnostic tool in medicine
(b)	X-rays	(ii)	Water purification
(c)	Microwave	(iii)	Communication, Radar
(d)	Infrared wave	(iv)	Improving visibility in foggy days

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(C) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(D) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Q63. MCQ (Single Correct Answer)

Open solution

The kinetic energy of emitted electron is E when the light incident on the metal has wavelength λ. To double the kinetic energy, the incident light must have wavelength:

(A) $\frac{hc}{E\lambda - hc}$

(B) $\frac{hc\lambda}{E\lambda + hc}$

(C) $\frac{h\lambda}{E\lambda + hc}$

(D) $\frac{hc\,\lambda}{E\lambda - hc}$

Q64. MCQ (Single Correct Answer)

Open solution

Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

(A) 3 : 4

(B) 4 : 3

(C) 1 : 4

(D) 4 : 1

Q65. MCQ (Single Correct Answer)

Open solution

A travelling microscope has 20 divisions per cm on the main scale while its vernier scale has total 50 divisions and 25 vernier scale divisions are equal to 24 main scale divisions, what is the least count of the travelling microscope?

(A) 0.001 cm

(B) 0.002 mm

(C) 0.002 cm

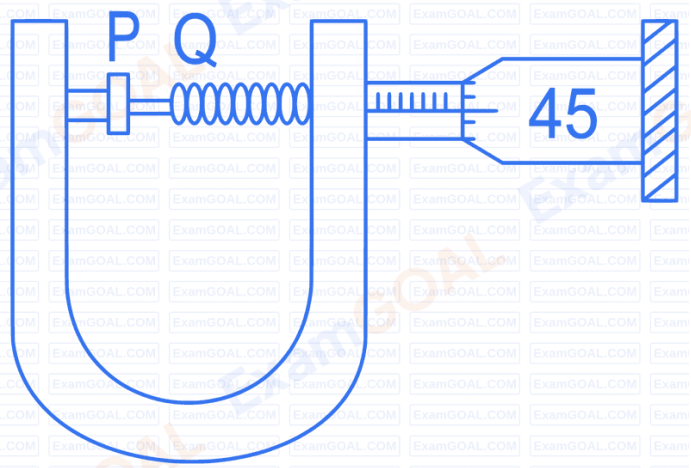
(D) 0.005 cm

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Q66. MCQ (Single Correct Answer)

Open solution

In an experiment to find out the diameter of wire using screw gauge, the following observations were noted :



(A) Screw moves $0.5\ \mathrm{mm}$ on main scale in one complete rotation

(B) Total divisions on circular scale $= 50$

(C) Main scale reading is $2.5\ \mathrm{mm}$

(D) 45th division of circular scale is in the pitch line

(E) Instrument has 0.03 mm negative error

Then the diameter of wire is :

(A) 2.92 mm **(B)** 2.54 mm **(C)** 2.98 mm **(D)** 3.45 mm

Q67. Numerical

Open solution

An object is projected in the air with initial velocity u at an angle θ. The projectile motion is such that the horizontal range R, is maximum. Another object is projected in the air with a horizontal range half of the range of first object. The initial velocity remains same in both the cases. The value of the angle of projection, at which the second object is projected, will be _____ degree.

Q68. Numerical

Open solution

If the acceleration due to gravity experienced by a point mass at a height h above the surface of earth is same as that of the acceleration due to gravity at a depth $\alpha h\,(h << R_e)$ from the earth surface. The value of α will be _____.

(use $R_e = 6400\ \mathrm{km}$)

Q69. Numerical

Open solution

The pressure P_1 and density d_1 of diatomic gas $\left(\gamma = \frac{7}{5}\right)$ changes suddenly to $P_2\,(> P_1)$ and d_2 respectively during an adiabatic process. The temperature of the gas increases and becomes _____ times of its initial temperature. (given $\frac{d_2}{d_1} = 32$)

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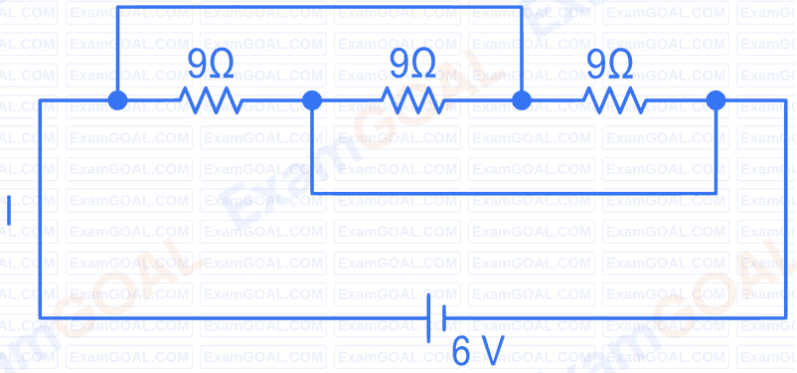
Q70. Numerical

One mole of a monoatomic gas is mixed with three moles of a diatomic gas. The molecular specific heat of mixture at constant volume is $\frac{\alpha^2}{4}\mathrm{R\,J/mol\,K}$; then the value of α will be _____. (Assume that the given diatomic gas has no vibrational mode).

Open solution

Q71. Numerical

The current I flowing through the given circuit will be _____A.



Open solution

Q72. Numerical

A closely wound circular coil of radius 5 cm produces a magnetic field of $37.68 \times 10^{-4}\,\mathrm{T}$ at its center. The current through the coil is _____A.

[Given, number of turns in the coil is 100 and $\pi = 3.14$]

Open solution

Q73. Numerical

Two light beams of intensities 4I and 9I interfere on a screen. The phase difference between these beams on the screen at point A is zero and at point B is π. The difference of resultant intensities, at the point A and B, will be _____ I.

Open solution

Q74. Numerical

A wire of length $314\,\mathrm{cm}$ carrying current of $14\,\mathrm{A}$ is bent to form a circle. The magnetic moment of the coil is _____ A $-\mathrm{m}^2$. [Given $\pi = 3.14$]

Open solution

Q75. Numerical

The X-Y plane be taken as the boundary between two transparent media M_1 and M_2. M_1 in $Z \geqslant 0$ has a refractive index of $\sqrt{2}$ and M_2 with $Z < 0$ has a refractive index of $\sqrt{3}$. A ray of light travelling in M_1 along the direction given by the vector $\vec{\mathrm{P}} = 4\sqrt{3}\hat{i} - 3\sqrt{3}\hat{j} - 5\hat{k}$, is incident on the plane of separation. The value of difference between the angle of incidence in M_1 and the angle of refraction in M_2 will be _____ degree.

Open solution

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Q76. Numerical

If the potential barrier across a p-n junction is $0.6\ \mathrm{V}$. Then the electric field intensity, in the depletion region having the width of $6 \times 10^{-6}\ \mathrm{m}$, will be _____ $\times 10^{5}\ \mathrm{N/C}$.

Open solution

Out of syllabus 14 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q77. MCQ (Single Correct Answer) · **Chemistry**

$100\ \mathrm{mL}$ of $5\%\ (\mathrm{w/v})$ solution of NaCl in water was prepared in $250\ \mathrm{mL}$ beaker. Albumin from the egg was poured into NaCl solution and stirred well. This resulted in a/an :

(A) Lyophilic sol (B) Lyophobic sol (C) Emulsion (D) Precipitate

Open solution

Q78. MCQ (Single Correct Answer) · **Chemistry**

In metallurgy the term "gangue" is used for :

(A) Contamination of undesired earthy materials.

(B) Contamination of metals, other than desired metal.

(C) Minerals which are naturally occurring in pure form

(D) Magnetic impurities in an ore.

Open solution

Q79. MCQ (Single Correct Answer) · **Chemistry**

Lithium nitrate and sodium nitrate, when heated separately, respectively, give :

(A) $\mathrm{LiNO_2}$ and $\mathrm{NaNO_2}$ (B) $\mathrm{Li_2O}$ and $\mathrm{Na_2O}$

(C) $\mathrm{Li_2O}$ and $\mathrm{NaNO_2}$ (D) $\mathrm{LiNO_2}$ and $\mathrm{Na_2O}$

Open solution

Q80. MCQ (Single Correct Answer) · **Chemistry**

Which among the following pairs has only herbicides?

(A) Aldrin and Dieldrin (B) Sodium chlorate and Aldrin

(C) Sodium arsinate and Dieldrin (D) Sodium chlorate and sodium arsinite

Open solution

Q81. MCQ (Single Correct Answer) · **Chemistry**

Which of the following compounds is an example of hypnotic drug?

(A) Seldane (B) Amytal (C) Aspartame (D) Prontosil

Open solution

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Q82. Numerical · **Chemistry**

Ionic radii of cation A^+ and anion B^- are 102 and 181 pm respectively. These ions are allowed to crystallize into an ionic solid. This crystal has cubic close packing for B^-. A^+ is present in all octahedral voids. The edge length of the unit cell of the crystal AB is _____ pm. (Nearest Integer)

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

If the foot of the perpendicular from the point $A(-1, 4, 3)$ on the plane $P : 2x + my + nz = 4$, is $\left(-2, \frac{7}{2}, \frac{3}{2}\right)$, then the distance of the point A from the plane P, measured parallel to a line with direction ratios $3, -1, -4$, is equal to :

(A) 1 (B) $\sqrt{26}$ (C) $2\sqrt{2}$ (D) $\sqrt{14}$

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a} = 3\hat{i} + \hat{j}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$. Let $\vec{c}$ be a vector satisfying $\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b} + \lambda\vec{c}$. If $\vec{b}$ and $\vec{c}$ are non-parallel, then the value of λ is :

(A) -5 (B) 5 (C) 1 (D) -1

Open solution

Q85. MCQ (Single Correct Answer) · **Mathematics**

The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot\alpha$ is equal to :

(A) $\frac{6}{5}$ (B) $\frac{9}{5}$ (C) $\frac{4}{3}$ (D) $\frac{7}{3}$

Open solution

Q86. MCQ (Single Correct Answer) · **Mathematics**

The statement $(p \wedge q) \Rightarrow (p \wedge r)$ is equivalent to :

(A) $q \Rightarrow (p \wedge r)$ (B) $p \Rightarrow (p \wedge r)$

(C) $(p \wedge r) \Rightarrow (p \wedge q)$ (D) $(p \wedge q) \Rightarrow r$

Open solution

Q87. Numerical · **Mathematics**

Let $S = \{\theta \in (0, 2\pi) : 7\cos^2\theta - 3\sin^2\theta - 2\cos^2 2\theta = 2\}$. Then, the sum of roots of all the equations $x^2 - 2\left(\tan^2\theta + \cot^2\theta\right)x + 6\sin^2\theta = 0, \theta \in S$, is _____.

Open solution

Q88. Numerical · **Mathematics**

Let a line with direction ratios $a, -4a, -7$ be perpendicular to the lines with direction ratios $3, -1, 2b$ and $b, a, -2$. If the point of intersection of the line $\frac{x+1}{a^2+b^2} = \frac{y-2}{a^2-b^2} = \frac{z}{1}$ and the plane $x - y + z = 0$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to _____.

Open solution

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Q89. Numerical · **Mathematics**

If $\frac{1}{2\times3\times4}+\frac{1}{3\times4\times5}+\frac{1}{4\times5\times6}+\ldots+\frac{1}{100\times101\times102}=\frac{k}{101}$, then 34 k is equal to _____.

Open solution

Q90. MCQ (Single Correct Answer) · **Physics**

Find the modulation index of an AM wave having $8\ \mathrm{V}$ variation where maximum amplitude of the AM wave is $9\ \mathrm{V}$.

(A) 0.8 (B) 0.5 (C) 0.2 (D) 0.1

Open solution

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Answer Key

Chemistry

1. **B**	2. **C**	3. **C**	4. **C**	5. **B**
6. **A**	7. **D**	8. **D**	9. **C**	10. **C**
11. **D**	12. **C**	13. **A**	14. **B**	15. **B**
16. **1000**	17. **548**	18. **54**	19. **1**	20. **282**
21. **40**	22. **1**	23. **3**	24. **2**	

Mathematics

25. **B**	26. **A**	27. **B**	28. **C**	29. **C**
30. **B**	31. **B**	32. **B**	33. **B**	34. **A**
35. **D**	36. **D**	37. **B**	38. **B**	39. **D**
40. **D**	41. **4**	42. **16**	43. **84**	44. **282**
45. **4**	46. **11**	47. **12**		

Physics

48. **D**	49. **B**	50. **B**	51. **A**	52. **C**
53. **A**	54. **A**	55. **A**	56. **A**	57. **C**
58. **A**	59. **B**	60. **C**	61. **C**	62. **B**
63. **B**	64. **A**	65. **C**	66. **C**	67. **15 or 75**
68. **2**	69. **4**	70. **3**	71. **2**	72. **3**
73. **24**	74. **11**	75. **15**	76. **1**	

Out of syllabus

77. **A**	78. **A**	79. **C**	80. **D**	81. **B**
82. **566**	83. **B**	84. **A**	85. **A**	86. **D**
87. **16**	88. **10**	89. **286**	90. **A**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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